Semester Project: Smart Parking Platform

Project Overview

WE ARE $oftware ¢orp. A fictitious software development studio with unlimited resources and budget.

The project involves the planning and management of a software system that helps drivers locate, reserve, and pay for parking spaces for parking garages in real time. The platform will be available through web and mobile applications and will provide parking availability information, navigation assistance, and management tools for parking operators.

The goal is to develop a project management plan for the creation of this system, including requirements gathering, stakeholder management, scheduling, budgeting, risk management, and team coordination.

Business Need

Drivers often spend significant time searching for available parking, leading to congestion, frustration, and lost productivity. Parking operators need better tools to monitor occupancy, manage pricing, and improve utilization of parking assets.

High-Level Requirements

The platform should provide:

- User registration and authentication
- Real-time parking space availability
- Interactive map displaying available parking locations
- Parking reservation capability
- Digital payment processing
- Reservation history and receipts
- Administrative dashboard for parking operators
- Occupancy reporting and analytics
- Notifications and alerts for users
- Integration with external mapping/navigation services
- Additional requirements should be identified through stakeholder analysis and requirements elicitation
- activities.

Stakeholders

- Drivers and vehicle owners
- Parking facility operators
- City transportation departments
- System administrators
- Finance and billing personnel
- Mobile application users
- Third-party payment providers

Development Teams

- Mobile Application Team
- Web Application Team
- Backend/API Team
- Mapping & Location Services Team
- Payment Integration Team
- Quality Assurance & Testing Team

Project Constraints

- Unlimited theoretical development budget
- Fixed semester timeline
- Dependence on third-party services and APIs
- Compliance with security and privacy requirements, city ordinance & laws

Research Component

You must perform a competitive analysis of existing parking solutions and identify strengths, weaknesses, and opportunities for differentiation.

Weekly updates to the project document

Every week's homework will start a part or section of the final project document. The weekly requirements are not 100% complete, and it is every student's responsibility to complete each component until the final project is due at the end of the semester.

Additionally, ensure that the final project document is well formatted, includes a header and a footer, a table of contents, and image captions.

Homework 1: Project Kickoff

1) RESEARCH: WHAT DOES ALREADY EXIST? ADD EXISTING SOFTWARE LANDSCAPE TO YOUR DOCUMENT, AND HOW YOUR APPLICATION WILL BE DIFFERENT FROM WHAT ALREADY EXISTS

2) START VISION AND SCOPE SECTION (THIS PART CAN AND WILL BE REFINED DURING THE COURSE) START WITH HOW THIS PROJECT WAS ACQUIRED, INTRODUCTION OF YOUR (FICTIONAL) COMPANY, AND ROUGH OVERVIEW OF WHAT THIS PROJECT IS ABOUT

3) DELIVER DRAFT FOR SRS, INCLUDING 15 USE CASES (MINIMUM) THIS SHOULD GROW OVER THE COURSE OF THIS SEMESTER TO COMPLETION

“To summarize:

- Ensure you have either a Word Document or a Markdown document in your GitHub Repository (public, not private).
- Make sure the document covers points 1, 2, and 3 of the assigned tasks, as listed in the end of last class's slides. The topic of our class project is in a separate project document outlined.
- Make sure you have created a 1-2 minute video as a stakeholder update what was done this week for the project
- Make sure you submit both the GitHub link, AND the video link on Canvas under Homework 1
- Make sure, if you have used AI, you have an accompanying ai use document submitted.”

Software Requirements Specification (SRS)

Project: <Project Name>

Version: 1.0

Date: 09.04.2026

1. Introduction

1.1 Purpose

This document defines the requirements for <Project Name>.

1.2 Scope

<Description of the scope of the project, and what is definitely out of scope>

2. Overall Description

- Users: <List the different groups of users that will be using the finished product>

- System Environment: Web-based / desktop / mobile application ?

- Constraints: Must run on specific hardware, must require specific authentication, etc

- Assumptions: Application has internet / intranet access, etc

3. Functional Requirements

FR1: The system shall allow users to register and log in securely.

FR2: The system shall allow users to add inventory items.

FR3: The system shall display daily sales reports.

FR4: The system shall automatically generate and email a weekly reports.

Etc

4. Non-Functional Requirements

NFR1: The system shall support at least 1,000 concurrent users.

NFR2: The system shall update inventory values within 1 second.

NFR3: The system shall provide 99.99% uptime.

Etc

5. Use Case Example

Title: Add new item to inventory

Actor: Warehouse Worker

Precondition: User is logged in, barcode scanner is installed

Steps:

1. User selects “Capture Inventory”

2. User scans new item’s barcode

3. User enters number of items to be shelved

4. User saves new inventory addition and verifies the updated number

Postcondition: captured item is available for purchase

Use Case Example

Use Case: Detailed, structured, often diagrammed.

User Story: Lightweight, agile, human-readable.

Example Use Case:

Actor: Salesperson

Goal: Sell item from inventory over the counter

Steps: See SRS above

Example User Story:

“As a salesperson, I need to scan an item at the

counter, accept payment, and transfer ownership of

the item.”

Acceptance Criteria:

• User can scan item barcode.

• System adds item to receipt.

• System verifies payment method.

• System updates inventory number of sold item.

Homework 2: Initial Documentation

Deliverables added to your Project Management Document(s):
1. Create WBS for Parking App (≥ 3 levels deep)
2. Draft timeline (including Gantt chart)

WBS Example

1. Frontend
1.1 Homepage
    Navigation
1.2 Product Search
    Autocomplete
1.3 Cart
    Cart actions
2. Backend
2.1 Authentication
    2FA w/ DUO
2.2 Inventory Management
    CRUD
2.3 Payment Gateway
    PayPal integration

Formula: FP = UFP × CAF
(Unadjusted FP x Complexity Adjustment Factor)

Homework 3: Work Backlog

1. Create backlog for Parking app:
- Login (min 5 items)
- UI for operator and user (min 15 items)
- Backend processes for operator and user (min 15 items)
- Reporting for operator and user (min 10 items)

2. Plan Sprint 1 backlog
Deliverable: Screenshot or exported board using either Jira or Trello, added to your document / repo

Homework 4: Project Risks and Team Communication

- **Identify and list at least 4 risks for each category: technical, schedule, financial, people.**
- **Build a Risk Register with at least 10 entries**
- **Create a Communication Plan: meeting cadence per team, reporting methods**

Homework 5: Roles and Budgeting

- **1) Roles & Resources: Define roles for your project. Provide (fictional) names & skill level assumptions and % allocations.**
- **2) Resource & Cost Plan: build a Bottom-Up or Top Down cost plan and a short budgeted cashflow (monthly). You can use the template tables.**
- **3) CREATE a RACI matrix for your project.**